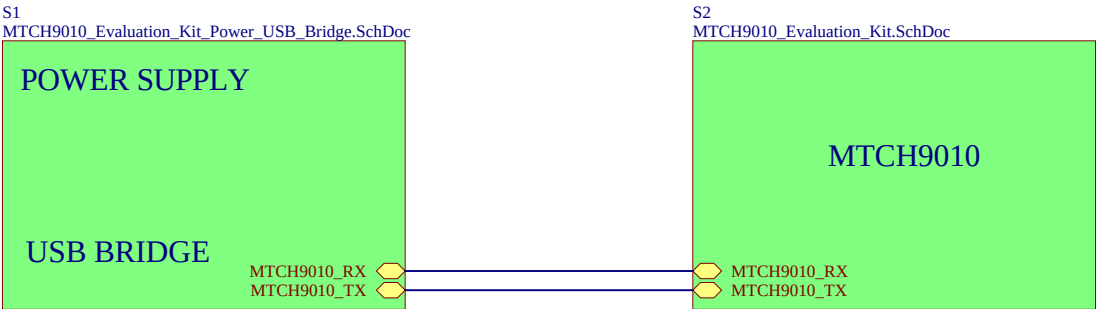


MTCH9010 Evaluation Kit



SCR1
Phillips Screw 1/4"

SCR2
Phillips Screw 1/4"

SCR3
Phillips Screw 1/4"

SCR4
Phillips Screw 1/4"

STANDOFF1
1/4" Standoff Nylon

STANDOFF2
1/4" Standoff Nylon

STANDOFF3
1/4" Standoff Nylon

STANDOFF4
1/4" Standoff Nylon

Test1
\$ >_

TEST, MTCH9010 Evaluation Kit

LABEL1

[ASSY# / REV]
SN: [SERIAL]
[DATE yyyy.mm.dd]

PCBA LABEL 18X6mm

S3
MTCH9010_Evaluation_Kit_Revision_History.SchDoc

Text in Silkscreen on PCB

Project Owner:
SLT

PCB Layout Contact:
SLT

PartNumber:
EV24U22A

Sheet Title
Top Level

Size
A3

Project Title
MTCH9010 Evaluation Kit

Variant: Default Assembly

Rev: 5

Date: 2025-04-14

Rev: 5

Sheet 1 of 4

File: MTCH9010_Evaluation_Kit_TopLevel.SchDoc

Microchip

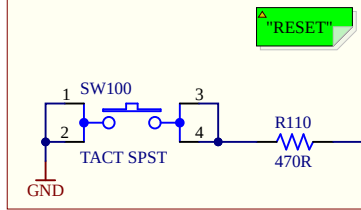
Designed with
Altium

Altium.com

CNANO Template Revision: 2.8

MTCH9010 Evaluation Kit

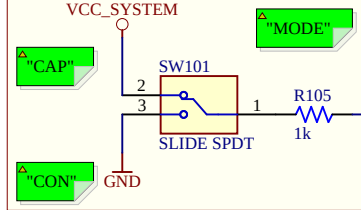
RESET BUTTON



RESET

Use the reset button to latch new board configuration.

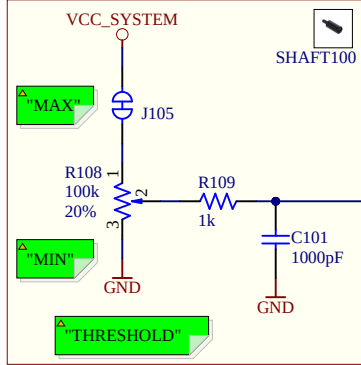
OPERATION MODE SELECTION



OPERATION_MODE

Select between capacitive or conductive operation. Pin setting can be overridden using Enhanced Configuration.

LIQUID DETECTION THRESHOLD

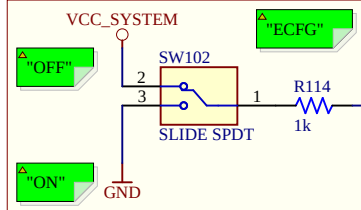


THRESHOLD

Detection threshold: lowest at MIN, highest at MAX. Threshold value can be overridden using Enhanced Configuration.

Detection threshold maximum can be adjusted by soldering a resistor to the cut-strap J105.

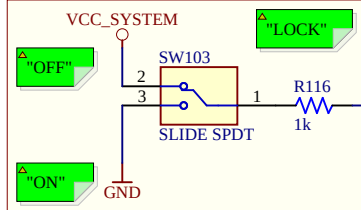
ENHANCED CONFIG ENABLE



ECFG_ENABLE

Enable or disable Enhanced Configuration Mode.

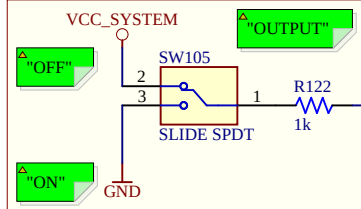
SYSTEM LOCK



SYSTEM_LOCK

Save configuration to NVM; requires Enhanced Configuration Mode.

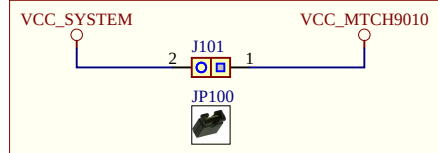
EXTENDED OUTPUT ENABLE



OUTPUT_ENABLE

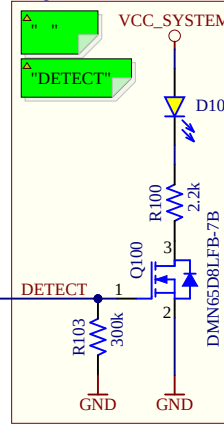
Enable or disable Extended Output. Pin setting can be overridden using Enhanced Configuration.

MTCH9010 CURRENT MEASUREMENT

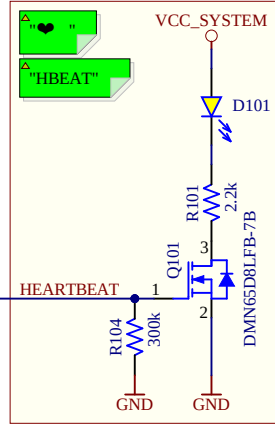


To measure the current consumption of the MTCH9010, remove JP100 and connect an ammeter to J101.

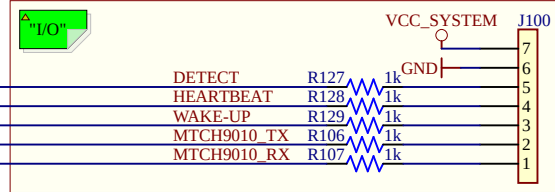
LIQUID DETECT LED



HEARTBEAT LED

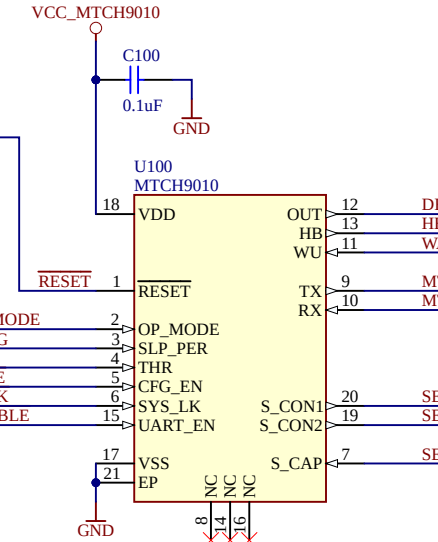


EXTERNAL CONTROL INTERFACE



The J100-7 pin can connect external circuitry to VCC_SYSTEM or receive external power. It is labeled "VDD" in silkscreen.

CAUTION: Power pins on J100 do not have polarity protection. Remove jumper from J201 when supplying external power.



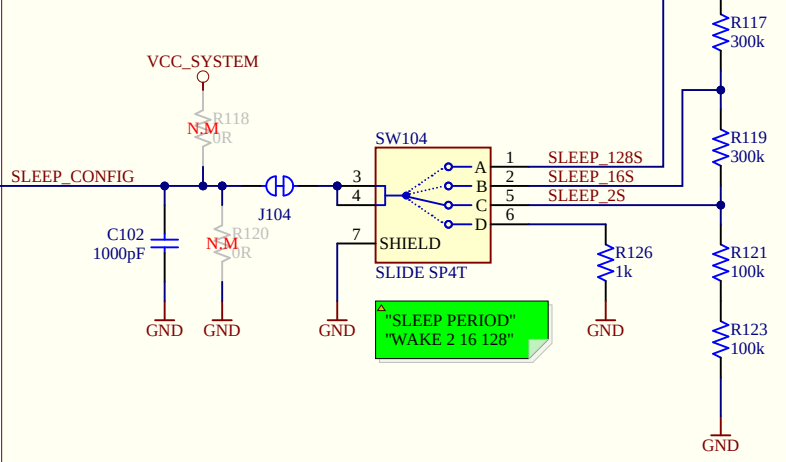
SLEEP PERIOD CONFIGURATION

The slide switch allows selecting a preset sleep period or waking the MTCH9010 by an external WAKE-UP signal.

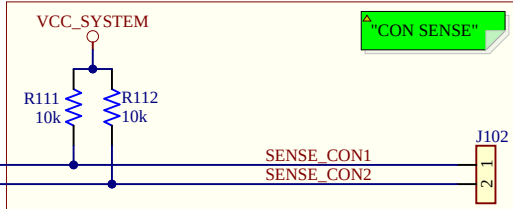
Preset sleep intervals: 2s, 16s, 128s

User-configurable sleep: Cut J104 and add suitable resistors to R118 and R120

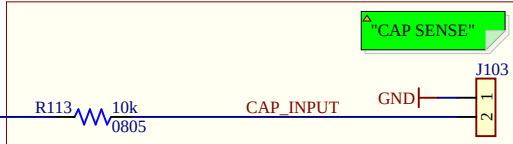
Sleep settings can be overridden using Enhanced Configuration.



CONDUCTIVE SENSOR HEADER

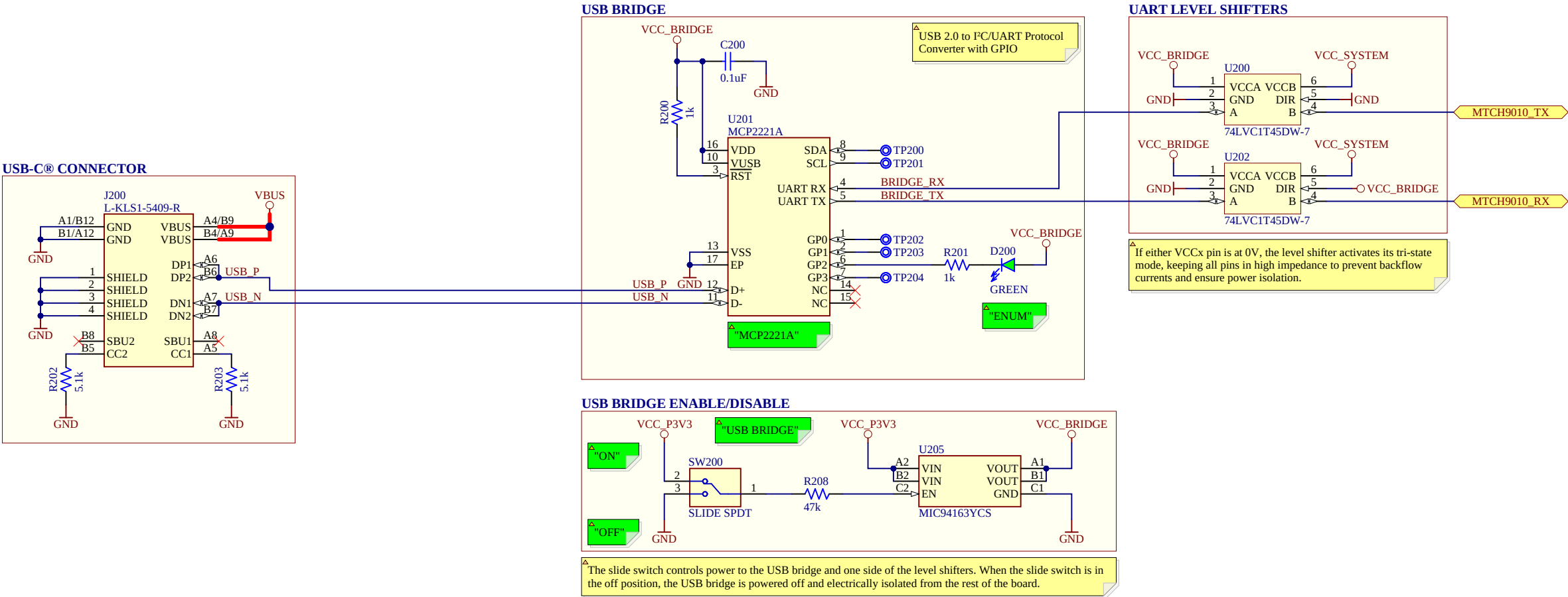


CAPACITIVE SENSOR HEADER

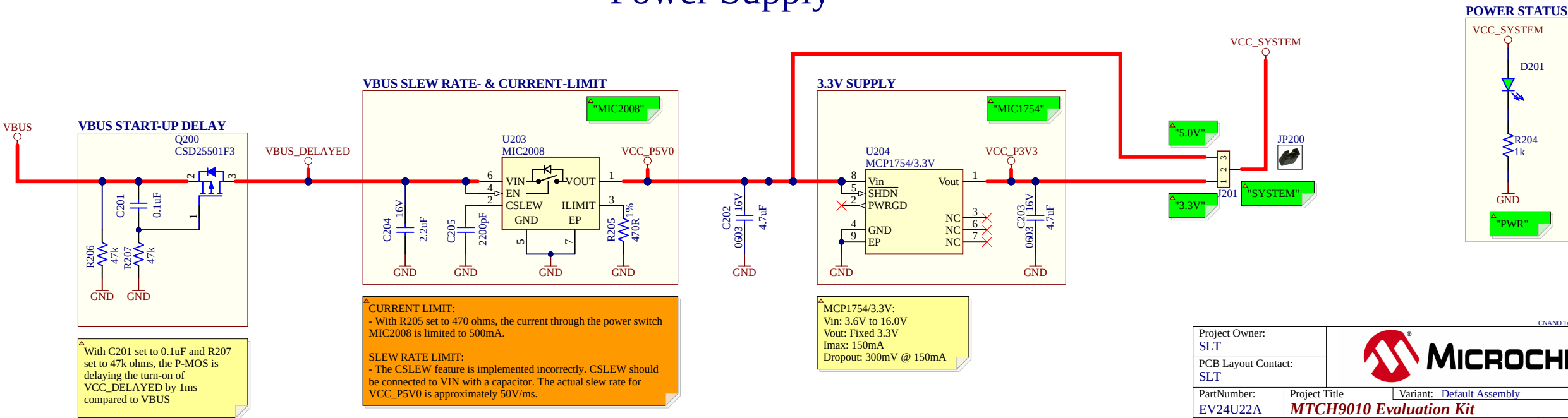


Project Owner: SLT		 MICROCHIP			
PCB Layout Contact: SLT					
PartNumber: EV24U22A		Project Title MTCH9010 Evaluation Kit		Variant: <u>Default Assembly</u>	
Sheet Title MTCH9010 Evaluation Kit					
Size A3	SCH #: 02-01228	Rev: 5	Date: 2025-04-14		 <u>Altium.com</u>
	PCB #: 04-12292	Rev: 5	Sheet 2 of 4		
File: MTCH9010_Evaluation_Kit.SchDoc					

USB Bridge



Power Supply



Revision History

PCB Assembly Rev 4:

Design Changes:

Initial Design

PCB:

PCB revision 4

PCB Assembly Rev 5:

Design Changes:

- Added capacitor and series resistor to reset circuit

- Changed Enhanced configuration switch connections

- Changed detection threshold potentiometer connections

- Added cut-strap to allow users to limit detection threshold maximum

PCB:

PCB revision 5

CNANO Template Revision: 2.8